Crime Analytics – Big Data Project Report

Project: Crime in Karnataka 2025 - End-to-End Big Data Pipeline (HDFS/PySpark + MySQL/SQLite + Streamlit)
Date: 2026-10-08

1. Objectives

- Build distributed pipeline: HDFS -> PySpark -> Spark SQL -> MySQL -> Streamlit.
- Extract/clean NCRB-style district-level crime data (Karnataka 2025).
- Compute rates, derive KPIs, store aggregated results for fast queries.
- Create interactive dashboard for analysis.

2. Data Sources & Validation

Source: Crime in Karnataka 2025 (Government of Karnataka Police Dept.). Extracted via pdfplumber to 26 CSVs.
Validation: 1521 trend rows, 39 districts, 3 years (2023–2025). Fact table + 13 normalized tables in DB.
Cross-checks: district_master_2025, state_trend_2011_2025, head-wise trends consistent.

3. Architecture

HDFS(raw/processed) -> PySpark(clean+transform) -> Spark SQL(analytics) -> MySQL/SQLite(aggregates) -> Streamlit(UI).
Files: scripts/hdfs_upload.sh, scripts/pyspark_crime_analytics.py, scripts/streamlit_app_sqlite.py, schema.sql, requirements.txt.

4. Results Summary

Fact rows: 1521
Districts: 39
Years: 2023-2025

5. Interpretation (Big Data)

Volume: structured dataset (pipeline mirrors distributed architecture). Variety: violent/women/property/missing/recidivism domains. Velocity: batch ETL. Veracity: cleaned/deduped/typed. Value: YoY, district ranking, domain share, recovery rates.

6. Data Sources Reference

Primary: Crime in Karnataka 2025 report (Karnataka Police). Extracted tables: district trends, master KPIs, state trend 2011–2025, IPC/SLL heads, violent heads x district, property, missing, recidivism.
Artifacts: CSVs (26), DB (SQLite), schema (MySQL), Streamlit dashboard, project scripts.